

1- What Does it Mean to Learn?

Exercises

A note before you start. Try each question yourself before turning to generative AI. Thinking through a problem, struggling with it for a while, and attempting your own answer are part of the learning process. Reading a polished AI-generated solution can easily create the impression that you understand something before you can explain or apply it yourself. If you get stuck, use generative AI to ask for a hint, a simpler question, or feedback on your reasoning rather than immediately asking for the solution. Make AI help you think, rather than do the thinking for you.

Exercises

1. Write the rules

You are given a 20×20 binary image containing a handwritten digit. Your task is to write a conventional computer program that decides whether the digit is a “3.”

Begin by proposing three or four rules that might help your program make this decision. You may use properties such as the number of black pixels, the presence of curves, the positions of pixels, or any other characteristics you consider useful.

Now consider what would happen if the digit were shifted a few pixels to the right, written more narrowly, rotated slightly, or drawn with a thicker stroke.

- (a) Which of your rules would still work?
- (b) How could you modify the rules to handle these variations?
- (c) What new difficulties would your modifications introduce?
- (d) At what point would you prefer to provide the computer with examples of handwritten digits instead of continuing to add rules?

Explain what this thought experiment reveals about the difference between explicit programming and learning from experience.

2. The knowledge engineer

Imagine that you are working in the 1980s and have been asked to build an expert system that helps diagnose faults in industrial machines. You have access to an experienced engineer who can answer questions for two hours each week.

After several months, your system contains hundreds of rules such as

IF the motor temperature is high AND the vibration is strong,
THEN inspect the bearings.

The factory then installs a new generation of machines, and several faults appear that the engineer has never encountered before.

Discuss the difficulties you would expect when extending the system. Which difficulties arise from obtaining knowledge from the expert, which arise from representing that knowledge as rules, and which arise when the system encounters unfamiliar situations? Explain why a learning-based approach might help, and identify one reason why expert knowledge could still remain useful.

3. Two collections of photographs

A company wants to build a system for recognizing ten species of birds. It can choose between two collections of images:

- Dataset A contains one million photographs taken by the same camera in the same wildlife park.
- Dataset B contains fifty thousand photographs taken by many people in different countries, seasons, weather conditions, backgrounds, and viewing angles.

The company can afford to use only one dataset.

Which would you choose? Justify your answer. In your discussion, distinguish the *amount* of experience from its *diversity* and *representativeness*. Can you imagine circumstances in which the other dataset would be preferable?

4. The mysterious shortcut

A system learns to distinguish photographs of wolves from photographs of dogs. It performs extremely well on the examples available to its developers. After deployment, its performance drops considerably.

A later investigation reveals that most wolf photographs used during learning contained snow, whereas most dog photographs did not.

- (a) Give a plausible explanation for the system's initial success.
- (b) Why might the same strategy fail on new photographs?
- (c) Suggest two new photographs that would help determine whether the system has learned something useful about wolves or has relied mainly on the background.

Relate your reasoning to the idea of generalization without assuming anything about the internal workings of the model.

5. Same world, different questions

A hospital has accumulated a large collection of chest X-rays. For some patients, doctors have also recorded a diagnosis.

Consider the following four projects:

- (a) predict whether a new X-ray shows pneumonia;
- (b) organize thousands of unlabeled X-rays into groups with similar visual characteristics;
- (c) remove part of each X-ray and ask a system to predict the missing region;
- (d) teach a robotic device to position an imaging sensor by rewarding good positioning and penalizing unsafe movements.

For each project, determine what form of experience is available to the learner and what kind of feedback, if any, tells it whether it is doing well. From this reasoning, identify the most appropriate learning paradigm. Focus on *why* the paradigms differ rather than simply naming them.

6. What exactly is being supervised?

A university wants to predict whether first-year students will successfully complete their degree. It has records containing previous grades, attendance, course choices, and the eventual outcome for students from the past ten years.

A colleague proposes three different objectives:

- (a) predict whether each student will graduate;
- (b) predict the student's final graduation grade;
- (c) discover groups of students with similar academic trajectories without specifying any desired outcome.

For each objective, identify what information would constitute the input and what, if anything, would constitute the desired output. Determine which problems are classification, regression, or neither. Explain why the same collection of observations can support several different learning problems.

7. The robot that learns in the wrong world

A warehouse robot acquires all of its experience in simulation before being placed in a real warehouse. The simulator contains shelves, boxes, workers, and moving vehicles, but every floor is perfectly flat, every box has the same friction, and the lighting never changes.

The robot performs extremely well in simulation and poorly in the real warehouse.

Suppose you are allowed to modify the simulator but cannot change the learning algorithm. What aspects of the simulated experience would you vary? Explain why exposing the robot to a wider range of conditions could improve its behavior in the real world. Would making the simulator visually more realistic necessarily solve the problem?

8. Design the experience

You have been asked to develop a machine learning system for detecting phishing websites. Before choosing any learning algorithm, your manager asks you to decide what experience should be collected.

Describe the data you would seek. Think about where the examples would come from, how varied they should be, how the correct outcomes could be established, and how quickly phishing strategies change over time. Then explain why collecting a very large dataset may still produce a poor learning system if the experience is badly chosen.

9. Four ingredients, one system

Choose a machine learning application that has *not* been used as an example in this chapter. Describe how you would begin designing it without naming a particular machine learning algorithm.

Your description should make clear:

- (a) what experience would be available;
- (b) what form the model's output should take;
- (c) what information the learning algorithm could use to improve the model;
- (d) how you would decide whether the resulting system performs well.

After answering, identify which of the four ingredients of a learning system each part of your description corresponds to.

10. **A claim worth questioning**

A company advertises its new product with the statement:

“Our AI has learned every possible situation from millions of examples.”

Write a short response explaining what is problematic about this statement. Your answer should use the ideas developed in this chapter without relying on mathematical terminology. In particular, discuss the relationship between experience, situations that have already been observed, and the situations the system may encounter after deployment.